AIORI-3 Hackathon 3 • Category: Cybersecurity

A3-PS019-TC539

PS-019: Personal-Data Discovery and Privacy Evidence Inventory

Proposed solution: **FieldLedger** — configurable, offline-first, evidence-backed discovery of candidate personal-data fields

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Status: planning stage
Proposed design. Nothing implemented or measured yet; no results claimed.

Abstract

Problem Description

- Personal-data fields accumulate across unindexed database tables and spreadsheets without clear documentation (PS-019).
- Teams need to find candidate personal-data fields and link them to declared business purpose and retention metadata.
- A discovery list alone is not enough: if purpose or retention is missing, the inventory cannot be reviewed meaningfully.
- Reviewers need to see why a field was flagged, so findings must be explainable and reviewable, not black-box labels.

What FieldLedger proposes to deliver

- Scanner for synthetic CSV files and one SQLite database
- Configurable detection rules with per-finding evidence
- Purpose and retention metadata mapping
- Missing-metadata and evidence-gap report
- Per-class precision and recall on a manually labelled set

Technical evidence for human review. No legal-compliance or lawful-basis decisions.

Background

Context for structured personal-data discovery

Structured discovery

Inspect columns in CSV files and SQL tables using column-name, content-pattern and schema signals. Output is a candidate, not proof.

Metadata governance

Organizations declare business purpose and retention per data asset. Missing declarations are an inventory gap.

Evidence-based review

Each finding records which rules fired and why, so a human can accept, reject or annotate it.

Evaluation

Detection is tested on a small manually labelled set; precision and recall reported per class.

Regulatory context [1][2]: India's DPDP Act 2023; DPDP Rules 2025 notified 13 Nov 2025 (G.S.R. 846(E)).

Technical discovery is not legal compliance. FieldLedger does not determine lawful basis or certify compliance.

Literature & RFC Review

Existing tools: what they document, and what remains unverified

Existing approach	Documented capability	Relevance to PS-019	Gap FieldLedger proposes / unverified
Presidio Structured [3]	presidio-structured: PII detection in tables and JSON; documented as alpha	Rule/recognizer-based, customizable detection	Purpose/retention mapping and gap reporting not described in page reviewed; not exhaustive
PIICatcher [4]	Column-name regex, plus regex/NLP on sampled column data; scans SQLite	Closest to name+content detection; this alone is not our novelty	Per-class evaluation and gap report not seen in README; maintenance status unverified
OpenMetadata auto-classification [5]	Column-name scanner plus sample-data entity recognition; PII tags	Same two signals inside a full catalog platform	Full catalog platform vs. lightweight offline tool; gap reporting not verified
DataHub [6]	Field-level tags, glossary terms, structured properties	Custom field properties could model retention (our reading)	Catalog; scanning and gap analysis not covered in page reviewed

Hypothesis to validate: offline workflow + evidence-backed gap report + per-class evaluation.

No RFC or CSV standard cited yet; none verified for this deck.

Proposed Solution

Synthetic CSV / SQLite sources → evidence-backed inventory and evaluation reports

1 Source ingestion

Read CSV files and one SQLite DB; raw values stay in memory

2 Configurable rules

Column-name, content-pattern and schema rules defined in config

3 Evidence records

Per field: rules fired, signals, match ratios, review status

4 Purpose & retention mapping

Join fields to declared metadata

5 Evidence-gap analysis

Flag missing purpose/retention and metadata conflicts

6 Inventory & reports

Structured inventory, gap report, per-class precision/recall

- Explainable: every finding shows its rules
- Configurable: rules and class taxonomy live in config
- Human review stays in the loop
- Measurable: labelled test set, TP/FP/FN per class
- Configurable evidence checklist (proposed): purpose, retention, evidence source, review status; format to confirm with mentor

Optimization Proposed

Comparison methodology: before and after

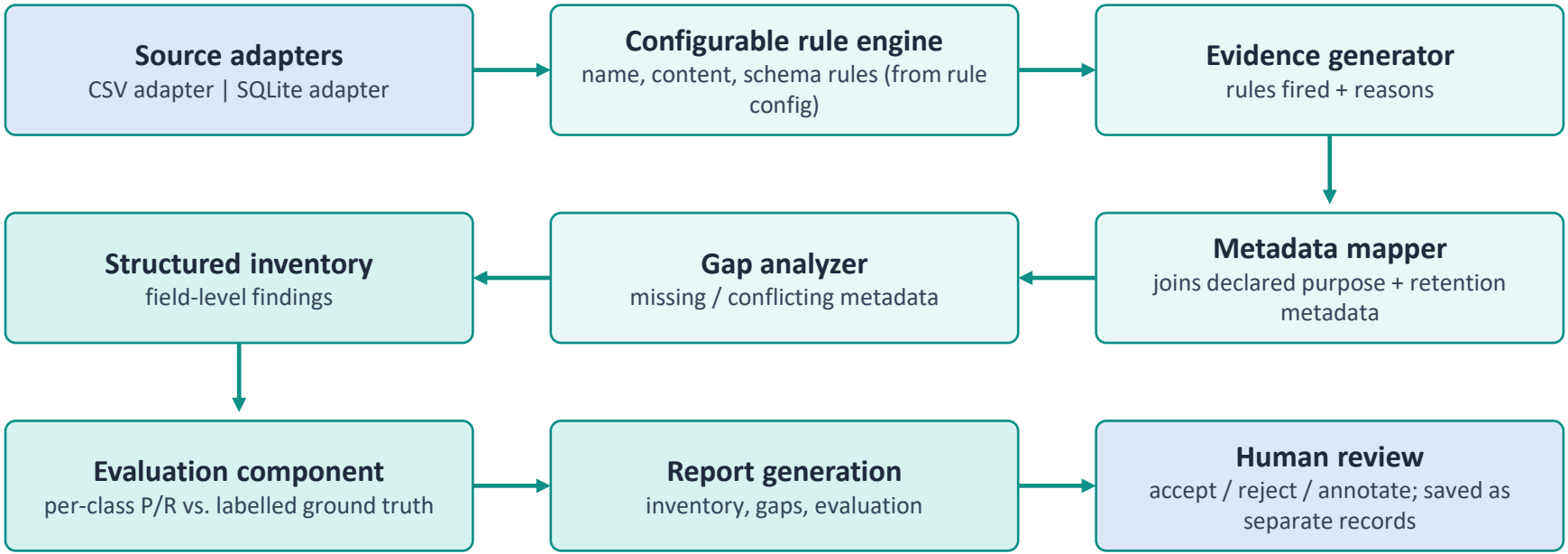
PROPOSED DESIGN. No measured results yet; values will be filled only after experiments run.

Approach	Signals	Output	What the experiment will test
A. Name-only baseline	Column-name rules	Flagged fields	Precision/recall per class; misses on ambiguous names
B. Content-pattern baseline	Value patterns	Flagged fields	False positives on look-alike values
C. Multi-signal (proposed)	Name + content + schema	Flagged fields with evidence	Change in per-class precision/recall vs. A and B
D. C + metadata mapping and gap report	C + purpose/retention declarations	Inventory + evidence-gap report	Do reviewers see missing/conflicting metadata that A–C cannot show? (seeded gaps)

Planned metrics: raw TP / FP / FN, per-class precision and recall, error analysis of false positives and false negatives.

Solution Architecture & Design

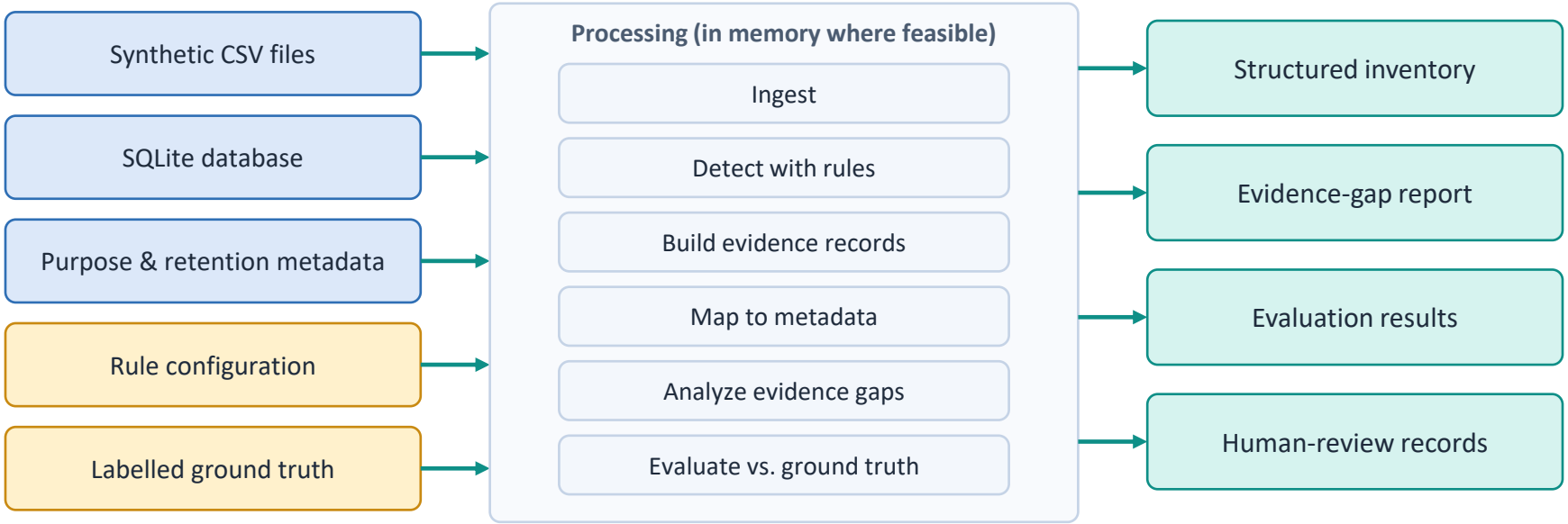
System Architecture: processing flow of FieldLedger components



External inputs: rule configuration → rule engine; declared purpose/retention metadata → metadata mapper; labelled ground truth → evaluation component. Planned design; not implemented. See Data Flow Diagram (slide 8).

Data Flow Diagram

Blue input • Amber configuration / ground truth • Grey in-memory processing • Teal output



Design intent: raw values are processed in memory and are not written to logs or reports; reports hold field names, rule IDs, counts.

Companion to the architecture on slide 7. Planned design; nothing implemented yet.

Deployment Timeline

Provisional roadmap. Dates to be set once the prototype deadline is confirmed.

1. Confirm official requirements and the five class definitions
2. Define data contracts and labelling rules
3. Build synthetic datasets and evaluation harness
4. Implement CSV and SQLite ingestion
5. Implement configurable detection and evidence generation
6. Implement metadata mapping and gap reporting
7. Integrate exports and minimal review interface
8. Test, evaluate, document, rehearse the demonstration

Current status: planning and initial repository setup. No milestone beyond that is complete.

Bibliography

Accessed October 14, 2026

[1] MeitY, Government of India. “Data Protection Framework.” (listed in PS-019) https://www.meity.gov.in/data-protection-framework

[2] MeitY. Gazette of India, Extraordinary, Part II, Sec. 3(i): G.S.R. 846(E), “Digital Personal Data Protection Rules, 2025,” 13 Nov 2025. https://www.meity.gov.in/static/uploads/2025/11/53450e6e5dc0bfa85ebd78686cadad39.pdf

[3] Microsoft. “Presidio structured.” Presidio documentation. https://microsoft.github.io/presidio/structured/

[4] Tokern. “PIICatcher” (GitHub repository README). https://github.com/tokern/piicatcher

[5] OpenMetadata. “Auto-Classification.” Documentation. https://docs.open-metadata.org/latest/how-to-guides/data-governance/classification/auto-classification

[6] DataHub. “SchemaField” (metadata model documentation). https://docs.datahub.com/docs/generated/metamodel/entities/schemafield-datahub

[7] AIORI-3. “PS-019: Personal-Data Discovery and Privacy Evidence Inventory” and “Proposed-structure-hackathon.pdf” (organizer documents).

Thank You

FieldLedger

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Repository: https://github.com/sudoyashvardhanvashistha/A3-PS019-TC539